Potassium: Fact Sheet for Consumers (original version)

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Table of Contents

What is potassium and what does it do?

Potassium is a mineral found in many foods. Your body needs potassium for almost everything it does, including proper kidney and heart function, muscle contraction, and nerve transmission.

How much potassium do I need?

The amount of potassium you need each day depends on your age and sex. Average daily recommended amounts are listed below in milligrams (mg).

Life Stage	Recommended Amount
Birth to 6 months	400 mg
Infants 7–12 months	860 mg
Children 1–3 years	2,000 mg
Children 4–8 years	2,300 mg
Children 9–13 years (boys)	2,500 mg
Children 9–13 years (girls)	2,300 mg
Teens 14–18 years (boys)	3,000 mg
Teens 14–18 years (girls)	2,300 mg
Adults 19+ years (men)	3,400 mg
Adults 19+ years (women)	2,600 mg
Pregnant teens	2,600 mg
Pregnant women	2,900 mg
Breastfeeding teens	2,500 mg
Breastfeeding women	2,800 mg

What foods provide potassium?

Potassium is found in many foods. You can get recommended amounts of potassium by eating a variety of foods, including the following:

- Fruits, such as dried apricots, prunes, raisins, orange juice, and bananas
- Vegetables, such as acorn squash, potatoes, spinach, tomatoes, and broccoli
- Lentils, kidney beans, soybeans, and nuts
- Milk and yogurt
- Meats, poultry, and fish

Salt substitutes

Potassium is an ingredient in many salt substitutes that people use to replace table salt. If you have kidney disease or if you take certain medications, these products could make your potassium levels too high. Talk to your health care provider before using salt substitutes.

What kinds of potassium dietary supplements are available?

Potassium is found in many multivitamin/mineral supplements and in supplements that contain only potassium. Potassium in supplements comes in many different forms—a common form is potassium chloride, but other forms used in supplements are potassium citrate, potassium phosphate, potassium aspartate, potassium bicarbonate, and potassium gluconate. Research has not shown that any form of potassium is better than the others. Most dietary supplements provide only small amounts of potassium, no more than 99 mg per serving.

Am I getting enough potassium?

The diets of many people in the United States provide less than recommended amounts of potassium. Even when food and dietary supplements are combined, total potassium intakes for most people are below recommended amounts.

Certain groups of people are more likely than others to have trouble getting enough potassium:

- People with inflammatory bowel disease (such as Crohn’s disease or ulcerative colitis)
- People who use certain medications (such as laxatives or some diuretics)
- People with pica (meaning that they eat things that aren’t food, such as clay)

What happens if I don't get enough potassium?

Getting too little potassium can increase blood pressure, deplete calcium in bones, and increase the risk of kidney stones.

Prolonged diarrhea or vomiting, laxative abuse, diuretic use, eating clay, heavy sweating, dialysis, or using certain medications can cause severe potassium deficiency. In this condition, called hypokalemia, blood levels of potassium are very low. Symptoms of hypokalemia include constipation, tiredness, muscle weakness, and not feeling well. More severe hypokalemia can cause increased urination, decreased brain function, high blood sugar levels, muscle paralysis, difficulty breathing, and irregular heartbeat. Severe hypokalemia can be life threatening.

What are some effects of potassium on health?

Scientists are studying potassium to understand how it affects health. Here are some examples of what this research has shown.

High blood pressure and stroke

High blood pressure is a major risk factor for coronary heart disease and stroke. People with low intakes of potassium have an increased risk of developing high blood pressure, especially if their diet is high in salt (sodium). Increasing the amount of potassium in your diet and decreasing the amount of sodium might help lower your blood pressure and reduce your risk of stroke.

Kidney stones

Getting too little potassium can deplete calcium from bones and increase the amount of calcium in urine. This calcium can form hard deposits (stones) in your kidneys, which can be very painful. Increasing the amount of potassium in your diet might reduce your risk of developing kidney stones.

Bone health

People who have high intakes of potassium from fruits and vegetables seem to have stronger bones. Eating more of these foods might improve your bone health by increasing bone mineral density (a measure of bone strength).

Blood sugar control and type 2 diabetes

Low intakes of potassium might increase blood sugar levels. Over time, this can increase the risk of developing insulin resistance and lead to type 2 diabetes. However, more research is needed to fully understand whether potassium intakes affect blood sugar levels and the risk of type 2 diabetes.

Can potassium be harmful?

Potassium from food and beverages has not been shown to cause any harm in healthy people who have normal kidney function. Excess potassium is eliminated in the urine.

However, people who have chronic kidney disease and those who use certain medications can develop abnormally high levels of potassium in their blood (a condition called hyperkalemia). Examples of these medications are angiotensin converting enzyme inhibitors, also known as ACE inhibitors, and potassium-sparing diuretics. Hyperkalemia can occur in these people even when they consume typical amounts of potassium from food.

Hyperkalemia can also develop in people with type 1 diabetes, congestive heart failure, liver disease, or adrenal insufficiency. Adrenal insufficiency is a condition in which the adrenal glands, located just above the kidneys, don't produce enough of certain hormones.

Even in healthy people, getting too much potassium from supplements or salt substitutes can cause hyperkalemia if they consume so much potassium that their bodies can't eliminate the excess.

People at risk of hyperkalemia should talk to their health care providers about how much potassium they can safely get from food, beverages, and supplements. The National Kidney Disease Education Program has information about food choices that can help lower potassium levels.

Does potassium interact with medications or other dietary supplements?

Yes, some medications may interact with potassium. Here are a few examples.

Angiotensin converting enzyme inhibitors and angiotensin receptor blockers

ACE inhibitors, such as benazepril (Lotensin), and angiotensin receptor blockers, such as losartan (Cozaar), are used to treat high blood pressure, heart disease, and kidney disease. They decrease the amount of potassium lost in the urine and can make potassium levels too high, especially in people who have kidney problems.

Potassium-sparing diuretics

Potassium-sparing diuretics, such as amiloride (Midamor) and spironolactone (Aldactone), are used to treat high blood pressure and congestive heart failure. These medications decrease the amount of potassium lost in the urine and can make potassium levels too high, especially in people who have kidney problems.

Loop and thiazide diuretics

Loop diuretics, such as furosemide (Lasix) and bumetanide (Bumex), and thiazide diuretics, such as chlorothiazide (Diuril) and metolazone (Zaroxolyn), are used to treat high blood pressure and edema. These medications increase the amount of potassium lost in the urine and can cause abnormally low levels of potassium.

Tell your doctor, pharmacist, and other health care providers about any dietary supplements and prescription or over-the-counter medicines you take. They can tell you if the dietary supplements might interact with your medicines or if the medicines might interfere with how your body absorbs, uses, or breaks down nutrients, such as potassium.

Potassium and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the *Dietary Guidelines for Americans*.

Where can I find out more about potassium?

- For general information about potassium
  - Office of Dietary Supplements (ODS) Health Professional Fact Sheet on Potassium
- For more information on food sources of potassium
  - USDA's FoodData Central
  - Nutrient list for potassium (listed by potassium content), USDA
- For more advice on choosing dietary supplements
  - ODS Frequently Asked Questions: Which brand(s) of dietary supplements should I purchase?
- For information about building a healthy dietary pattern
  - *Dietary Guidelines for Americans*

Glossary

absorption
: In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

ACE inhibitor
: Angiotensin-converting enzyme inhibitor. A medicine used to treat high blood pressure, stroke, heart attack, diabetes, and kidney problems

angiotensin receptor blocker
: A medication that relaxes and opens up blood vessels. This lowers blood pressure so the heart doesn't have to work as hard to pump blood.

blood sugar
: The main source of energy used by the body's cells. Blood sugar comes from food and is made by the liver, and is carried to the cells through the bloodstream. Also called blood glucose.

bone density
: A measurement of bone mass and an indicator of bone strength and health. Also called bone mineral density.

calcium
: A mineral found throughout the body. Calcium is needed for healthy bones and teeth, for nerves and enzymes to function properly, and for blood clotting. Calcium is found in some foods, including milk, yogurt, and cheese, and in Chinese cabbage, kale, broccoli and fortified foods, such as many drinks, tofu, and cereals.

chronic
: Happening for a long time, persistently, or repeatedly.

constipation
: A condition in which stool becomes hard, dry, and difficult to pass and bowel movements happen infrequently. Other symptoms may include painful bowel movements and feeling bloated, uncomfortable, and sluggish.

consume
: To eat or drink.

coronary heart disease
: A disease in which the blood vessels (coronary arteries) that carry blood and oxygen to the heart are narrowed or blocked, which can cause chest pain, shortness of breath, and heart attack. It is usually caused by a build-up of fat and cholesterol deposits inside the arteries (atherosclerosis).

Crohn's disease
: A long-lasting (chronic) disease that causes severe irritation in the gastrointestinal tract. It usually affects the lower small intestine (called the ileum) or the colon, but it can affect any part of the digestive tract from the mouth to the anus. It is painful, causing severe watery or bloody diarrhea, and may lead to life-threatening complications. Crohn's disease is a form of inflammatory bowel disease.

deficiency
: An amount that is not enough; a shortage.

diabetes
: A disease in which blood sugar (glucose) levels are high because the body is unable to use glucose properly. Diabetes occurs when the body does not make enough insulin, which helps the cells use glucose, or when the body no longer responds to insulin.

dialysis
: The process of filtering the blood when the kidneys are not able to cleanse it.

diarrhea
: Loose, watery stools.

dietary fiber
: A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Guidelines for Americans
: Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

dietary supplement
: A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

diuretic
: A drug or other substance that increases the amount of urine made by the body.

edema
: Swelling caused by excess fluid in the body. Edema often affects the hands, arms, feet, ankles, legs, hands, and arms.

enzyme
: A protein that speeds up chemical reactions in the body.

fortified

When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

gland
: A small organ that makes and releases a substance such as sweat, tears, saliva, milk, a hormone, or substances that aid in digestion.

heart failure
: A condition in which the heart is unable to pump the amount of blood needed by the body. It is caused by high blood pressure, heart attack, and other disorders of the heart or blood vessels. Also called congestive heart failure.

high blood pressure
: A blood pressure measurement of 140/90 mmHg (millimeters of mercury) or higher is considered high blood pressure (hypertension). Blood pressure is the force of blood pushing against the walls of the arteries. Blood pressure measurements are written as two numbers, for example 120/80. The first number (the systolic pressure) measures the pressure when the heart beats and pumps out blood into the arteries. The second number (the diastolic pressure) measures the pressure when the heart is at rest between beats. High blood pressure is a condition that occurs when a person's blood pressure often measures above 140/90 or regularly stays at that level or higher. This condition usually has no symptoms but can be life-threatening. It damages the arteries and increases the chance of stroke, heart attack, kidney failure, and blindness. Also called hypertension.

hormone
: A group of chemicals made by glands in the body. Hormones circulate in the bloodstream and control the actions of certain cells or organs. Some hormones can also be manufactured.

infant
: A child younger than 12 months old.

inflammatory bowel disease
: IBD. Long-lasting (chronic) problems that cause irritation and ulcers in the digestive tract. The most common disorders are ulcerative colitis and Crohn's disease.

ingredient
: In a dietary supplement, an ingredient is a component of the product, such as the main nutrient (vitamin, mineral, herb, amino acid, or enzyme) or any binder, color, filler flavor, or sweetener. In herbal supplements, the common name and Latin name (the genus and species) of the plant is given in the ingredient list. On a dietary supplement label, the ingredients are listed by weight, with the ingredient used in the largest amount first on the list and the ingredient used in the least amount at the end of the list.

insulin resistance
: A condition in which glucose (blood sugar) cannot be absorbed by the cells and used for energy. Instead, glucose builds up in the blood and the body produces more and more insulin (which normally would help glucose get into the cells), resulting in abnormally high blood levels of both glucose and insulin. This can lead to pre-diabetes, type 2 diabetes, and other serious health problems.

interaction
: A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

kidney
: One of two organs that remove waste from the blood (as urine). The kidneys also make erythropoietin (a substance that stimulates red blood cell production) and help regulate blood pressure. The kidneys are located near the back under the lower ribs.

kidney disease
: A condition that lessens the ability of the kidneys to filter wastes from the blood, keep blood chemical levels balanced, and make certain hormones. The two most common causes of kidney disease are diabetes and high blood pressure. People with long-term kidney disease may need dialysis or a kidney transplant to stay alive.

kidney stone
: A “pebble” that forms in a kidney from salts and minerals in the urine. A small kidney stone is able to pass out of the body; a large stone may block the urinary tract and require medical help.

laxative
: A substance that moves the bowels and relieves constipation.

liver
: A large organ located in the right upper abdomen. It stores nutrients that come from food, makes chemicals needed by the body, and breaks down some medicines and harmful substances so they can be removed from the body.

loop diuretic
: Medication that is used to treat fluid build-up in the body and congestive heart failure. It may also be used to treat high blood pressure. Loop diuretics increase the amount of water that the body loses as urine.

milligram
: mg. A measure of weight. It is a metric unit of mass equal to 0.001 gram (it weighs 28,000 times less than an ounce).

mineral
: In nutrition, an inorganic substance found in the earth that is required to maintain health.

neuron
: A nerve cell. Neurons send chemical and electrical messages throughout the nervous system that direct the body to function, move, think, and have emotions.

nutrient
: A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

Office of Dietary Supplements
: ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

pharmacist
: A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

pica
: An eating disorder in which a person eats things that are not food, such as dirt, clay, paint flakes, sand, hair, or paper. Pica is more common in young children and in people with brain injuries or developmental disabilities.

potassium
: A mineral that helps the body's nerves to function, muscles to move, and heart to beat. Potassium helps balance some of the harmful effects of salt on blood pressure.

poultry
: Birds that are raised for eggs or meat, including chickens, turkeys, ducks, and geese.

prescription
: A written order from a health care provider for medicine, therapy, or tests.

risk
: The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

risk factor
: Something that may increase the chance of developing a disease. For example, a diet that is low in calcium and vitamin D is a risk factor for osteoporosis; smoking is a risk factor for lung cancer.

soy
: A plant that produces beans used in many food products. Soy products contain isoflavones (estrogen-like substances) that are being studied in the prevention of cancer, hot flashes that occur with menopause, and osteoporosis (loss of bone density). Also called soya and soybean. Latin name: *Glycine max*.

stroke
: A loss of blood flow to part of the brain. Strokes are caused by blood clots or broken blood vessels in the brain, and result in damage to a section of brain tissue. Symptoms include dizziness, numbness, weakness on one side of the body, and problems with talking or understanding language. The chance (risk) of stroke is increased by high blood pressure, older age, smoking, diabetes, high cholesterol, heart disease, a family history of stroke, and a build-up of fatty material inside the coronary arteries (atherosclerosis). See also NIH publication: Know Stroke. Know the Signs. Act in Time. http://www.ninds.nih.gov/disorders/stroke/knowstroke.htm

supplement

A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

symptom
: A feeling of sickness that an individual can sense, but that cannot be measured by a healthcare professional. Examples include headache, tiredness, stomach ache, depression, and pain.

thiazide diuretic
: A drug used in the treatment of high blood pressure and swelling caused by excess fluid in body tissues (edema). It increases the amount of urine made by the body.

treat
: To care for a patient with a disease by using medicine, surgery, or other approaches.

ulcerative colitis
: Chronic inflammation of the colon that causes ulcers to form in its lining. This condition is marked by abdominal pain, cramps, and loose discharges of pus, blood, and mucus from the bowel.

urine
: Excess liquids and wastes that have been filtered from the blood by the kidneys, stored in the bladder, and removed from the body through the urethra (the tube that carries urine from the bladder to outside the body).

US Department of Agriculture
: USDA promotes America's health through food and nutrition, and advances the science of nutrition by monitoring food and nutrient consumption and updating nutrient requirements and food composition data. USDA is responsible for food safety, improving nutrition and health by providing food assistance and nutrition education, expanding markets for agricultural products, managing and protecting US public and private lands, and providing financial programs to improve the economy and quality of rural American life.

vitamin
: A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

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